

# NAZIRU HALILU

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## RESEARCH INTERESTS

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Computer vision and geospatial machine learning for digital agriculture; self-supervised and contrastive learning for agronomic imaging; radiometric harmonization and vegetation indexing from UAV and satellite imagery; geostatistics and precision agriculture mapping (regression kriging, singularity indices); hybrid physics-informed deep learning for hydrological and environmental process modelling; high-throughput field phenotyping; closed-loop sensing and control for smart irrigation and greenhouse automation.

## EDUCATION

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**Erasmus Mundus Joint Master's Degree (EMJMD), Cutting-Edge Technologies for Sustainable Agriculture** 2025 to expected 2027

*Public University of Navarre, Spain · UTAD, Portugal · Agricultural University of Athens, Greece*

**Bachelor of Engineering (B.Eng.), Agricultural and Bioresources Engineering, First Class Honours** 2019 to 2024

*Ahmadu Bello University, Zaria, Nigeria · CGPA 4.77/5.00, top 1% of graduating cohort*

**National Diploma, Agricultural Engineering Technology, Distinction** 2016 to 2018

*Ahmadu Bello University, Zaria, Nigeria · CGPA 3.78/4.00, top 1% of graduating cohort*

## RESEARCH EXPERIENCE

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**EMJMD Research Scholar (Mobility Track)** Sep 2026 to Jan 2027

*Agricultural University of Athens, Greece*

Term 3 focus: Efficient water use, smart irrigation (30 ECTS)

- Continued graduate-level research training in smart design and management of irrigation systems.
- Accomplished a complete farm-level irrigation and crop-growth decision-support tool, as measured by successful 4-step simulation (canopy, transpiration, biomass, yield) across five smallholder crops, by engineering NaCROP, a modular Python pipeline with a Tkinter desktop GUI for evapotranspiration, soil-water balance, and irrigation scheduling around Zaria, Nigeria. [Data](#)

**Summer Intern, Computer Vision Group** Jun 2026 to Aug 2026

*Dept. of Computer Engineering, Ahmadu Bello University, Zaria*

- Accomplished cross-condition transfer of crop-disease recognition models, as measured by 64.4% classification accuracy (macro AUC 0.882 against a 5.3% chance baseline) across 19 disease classes on 1,543 field images, by extending the Geometry-Agnostic Contrastive Learning (GACL) framework with four new geometry-invariant neural components, formalized as a first-author paper under review at Biosystems Engineering (Elsevier). [Data](#)

**EMJMD Research Scholar (Mobility Track)** Feb 2026 to Jun 2026

*University of Trás-os-Montes and Alto Douro (UTAD), Portugal*

Term 2 focus: Precision agriculture for sustainable crop production (30 ECTS)

- Accomplished cross-system precision-agriculture prediction and input-cost reduction, as measured by R-squared between 0.978 and 0.995 for grape ripening and maize yield regressions and a 2.69-tonne cut in lime demand, by integrating GIS, remote sensing, and machine learning pipelines across more than 65 hectares of Portuguese trial sites spanning three contrasting cropping systems. [Data](#)
- Accomplished more reliable soil-nutrient and yield mapping, as measured by a roughly 20% reduction in residual kriging uncertainty across 17,864 samples on a 16.85-hectare maize field (Moran's I between 0.72 and 0.98,  $p < 0.001$ ), by developing Multitask Singularity Regression Kriging (MSRK), coupling multifractal singularity indices with shared random forest models. [Data](#)
- Accomplished predictive harvest-date forecasting for grape ripening, as measured by R-squared = 0.9784 and a maturity forecast of 25.5 to 27 °Bx, by building an NDVI and logarithmic regression model from field must samples across five sampling dates at a 6-hectare UTAD vineyard, Quinta de Nossa Senhora de Lurdes, Vila Real. [Data](#)
- Accomplished a costed, spatially targeted pest-management programme, as measured by a sized pheromone diffuser plan (EUR 2,670.6 to 3,052.1) across 11 grape-variety blocks, by running terrain characterisation and a 25-trap spatial monitoring network for grape moth (*Lobesia botrana*) across a 42.97-hectare vineyard estate in the Douro Demarcated Region. [Data](#)

## EMJMD Research Scholar (Mobility Track)

Sep 2025 to 31 Jan 2026

Public University of Navarre, Spain

Term 1 focus: Technological bases for agricultural sustainability (30 ECTS)

- Accomplished real-time, field-deployable crop-health screening, as measured by 33.8% balanced accuracy against a 4.55% chance baseline across 22 disease classes on 1,543 validation images, by building a fourteen-module open-source pipeline for RGB/multispectral vegetation indexing, GLCM texture analysis, and CNN/Faster R-CNN inference. [Data](#)
- Accomplished radiometrically consistent multi-temporal vegetation monitoring, as measured by a 39% and 31% reduction in temporal NDVI variance for water and rooftop reference targets respectively (R-squared of at least 0.999 fidelity to raw NDVI), by designing STADOS, a five-stage adaptive dark-object subtraction framework validated on real Sentinel-2 imagery over Portuguese corn parcels. [Data](#)
- Accomplished accurate land-suitability mapping for livestock slurry application, as measured by RMSE = 7.78 and R-squared = 0.77 on a held-out set of 127 sites, outperforming three competing kriging and deep-learning alternatives, by building a two-layer GIS and regression-kriging pipeline across 550 spatially cross-validated observations over 14.35 square kilometres near Tudela, Spain. [Data](#)
- Accomplished a 47.8% reduction in annual soil loss and Nash-Sutcliffe model efficiency between 0.88 and 0.94, as measured against a vegetative erosion-control strategy for an agricultural gully system, by building a Bayesian-grounded digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models (hold-out R-squared = 0.90). [Data](#)
- Built a six-notebook vector and raster GIS, Sentinel-1/2 remote sensing, and classification course package covering supervised and unsupervised land-cover classification in Python and Google Earth Engine. [Data](#)

## Undergraduate Project Assistantships

Mar 2024 to Mar 2025

Depts. of Agricultural and Bioresources Engineering and Animal Science, Ahmadu Bello University

- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects spanning water and irrigation systems (hydroponics, greywater reuse, spatial infiltration, stream water quality), soil and erosion assessment (soil erodibility, rainfall erosivity), and postharvest and animal systems (evaporative cooling, fish pond systems, Moringa-based animal nutrition).

## Project Assistant (Remote Data Lead), Smart Greenhouse Automation

2024 to 2025

IAR-ABU, Nigeria

- Automated greenhouse environmental control by engineering adaptive irrigation-scheduling algorithms in Python that fused real-time soil-moisture and microclimate telemetry, serving as remote data lead for the full sensor-to-actuation pipeline.

## Student Research Assistant, Precision Crop Systems

2022 to 2024

NAERLS, Nigeria

- Enabled high-throughput field phenotyping by training and validating CNN-based crop-stress detection models in TensorFlow that reached 78% classification accuracy on multispectral UAV imagery, and by automating UAV flight-path planning for data acquisition.

## Assistant Farm Manager (concurrent, part-time)

2022 to 2024

Yankari Nasarawa Farms, Nigeria

- Sustained animal welfare and operational efficiency across a commercial poultry facility by managing automated climate-control and water-dosing systems and day-to-day operations alongside concurrent research assistant duties.

## Project Assistant, Smart Irrigation and Water Systems

2021 to 2022

BOMO-Basin Project, Nigeria

- Automated water delivery across a one-hectare smart drip-irrigation array by implementing closed-loop control logic and quantifying feedback-loop performance linking real-time soil-moisture data to scheduling.

## Research Intern

2017 to 2020

NCAM, Ilorin and IAR, Zaria, Nigeria

- Benchmarked mechanization efficiency of locally engineered machinery by building Python data-logging pipelines at NCAM, and supported baseline hydrological research at IAR by collecting and documenting soil-profile and environmental data.

## Virtual Secondary School Teacher

2024 to present

Remote, Nigeria

- Delivered ongoing remote instruction to secondary school students alongside graduate studies.

## Founder, Nhalilu's Biogas Plant and Sales Company

Registered with the Small and Medium Enterprises Development Agency of Nigeria (SMEDAN), Bioprocess Engineering

- Founded and registered a biogas production and sales venture, applying bioprocess engineering principles learned through academic research to a working commercial operation.

[View SMEDAN Certificate of Registration](#)

## PUBLICATIONS

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### Peer-Reviewed

- **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *International Journal of Research and Scientific Innovation (IJRSI)*, 13(7), 2920–2950. [DOI](#) | [Data](#)
- **Halilu, N.**, Sulaiman, J. (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951–2989. [DOI](#) | [Data](#)
- **Halilu, N.**, Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. *IJRSI*, 13(7), 2990–3020. [DOI](#)
- Ismail, K., **Halilu, N.** (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. African Conf. on Precision Agriculture (AfCPA)*, 2(1), 100–120.
- Auwal, M., Yusuf, A., **Halilu, N.** (2020). Physio-chemical and bacteriological quality of water for irrigation purpose (A case study of Shika Dam). *Nigerian Research Journal of Engineering and Environmental Sciences*, 5(1), 198–204.

### Submitted / Under Review

- **Halilu, N.**, Sulaiman, J. Geometry-Agnostic Contrastive Learning (GACL): A Theoretical Framework and Baseline Evaluation Protocol for Agronomic Imaging. *Biosystems Engineering* (Elsevier). Under review. [DOI](#) | [Data](#)
- **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (*Ipomoea* spp.). *Environmental Monitoring and Assessment* (Springer Nature). Under review. [DOI](#) | [Data](#)
- **Halilu, N.**, Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy* (Elsevier). Under review. [DOI](#) | [Data](#)

### Preprints

- **Halilu, N.** (2026). Multitask Singularity Regression Kriging for Spatio-Temporal Soil NPK Crop Stress Yield-Prediction. *Zenodo Preprint* (v1.0.1). [DOI](#) | [Data](#)
- **Halilu, N.** (2026). STADOS: Spectral-Temporal Adaptive Dark Object Subtraction for Radiometrically Harmonized Multi-Temporal Agricultural Vegetation Monitoring. *Zenodo*. [DOI](#) | [Data](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time-Series Modelling and Control Systems in Precision Irrigation. *Zenodo*. [DOI](#)

## AWARDS AND HONORS

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- Erasmus Mundus Joint Master Scholarship, European Union (2025)
- Best Project Award, ABU Senate (2024)
- Best Graduating Student, Agricultural & Bioresources Engineering (2024)
- Nigerian Federal Government Award for Academic Excellence (2024)
- Best Undergraduate Research Paper Award, ABU (2024)
- PTDF Undergraduate Fellow (2023)

## TEACHING AND MENTORSHIP

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- Undergraduate Course Tutor, ABU Zaria (NAMET), 2020–2024: Differential Equations & Transforms, Irrigation System Design, Computer Applications in Agric. Engineering.
- Deputy Coordinator, KOJYAF, 2020–2025: Mentorship programs preparing secondary students for JAMB entrance examinations.
- Virtual Secondary School Teacher, 2024–present: Remote instruction in Physics and Mathematics.

## PROFESSIONAL MEMBERSHIPS

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American Society of Agricultural and Biological Engineers (ASABE) · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN

## RESEARCH PROJECTS AND OPEN-SOURCE REPOSITORIES

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- Geometry-Agnostic Contrastive Learning (GACL), <https://github.com/halilunaziru73-creator/Geometry-Agnostic-Contrastive-Learning-GACL>
- STADOS: Spectral-Temporal Adaptive Dark Object Subtraction, <https://github.com/halilunaziru73-creator/STADOS-Spectral-Temporal-Adaptive-Dark-Object-Subtraction>
- Multitask Singularity Regression Kriging (MSRK), <https://github.com/halilunaziru73-creator/Multitask-Singularity-Regression-Kriging-for-Spatio-Temporal-Soil-NPK-Crop-Stress-Yield-Prediction>
- Digital Twin for Gully Biocontrol Using Morning Glory, <https://github.com/halilunaziru73-creator/Digital-Twin-for-the-Evaluation-of-Experimental-Gully-Biocontrol-Using-Morning-Glory-Ipomoea-spp>
- Hybrid CNN-BiLSTM-Attention Sediment Transport Framework, <https://github.com/halilunaziru73-creator/Hybrid-CNN-BiLSTM-Attention-Sediment-Transport-Agricultural-Gully-System>
- Real-Time RGB Proxy Vegetation Indexing and Texture Analysis, <https://github.com/halilunaziru73-creator/Real-Time-RGB-Proxy-Vegetation-Indexing-and-Texture-Analysis-for-UAV-and-Handheld-Crop-Imagery>
- Operationalizing GIS and ML across Contrasting Cropping Systems, <https://github.com/halilunaziru73-creator/Operationalizing-GIS-and-Machine-Learning-across-Contrasting-Cropping-Systems>
- Geospatial Data Analysis (six-notebook GIS/remote sensing course), <https://github.com/halilunaziru73-creator/Geospatial-data-analysis>
- GIS-Based Livestock Slurry Suitability Delineation, <https://github.com/halilunaziru73-creator/GIS-based-delineation-of-areas-suitable-for-livestock-slurry-application>
- GIS Monitoring of Grape Moth (*Lobesia botrana*), <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Grape-Moth-Vineyard-Pest>
- GIS Monitoring of Vineyard Ripening (Brix), <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Vineyard-Ripening-Brix>
- NaCROP: Crop ET and Irrigation Decision-Support Pipeline, <https://github.com/halilunaziru73-creator/NaCROP>
- Integrated Decision-Support Model for Dairy Processing, <https://github.com/halilunaziru73-creator/Integrated-Decision-Support-Model-for-Dairy-Processing-under-Milk-and-Market-Variability>
- Full repository index, <https://github.com/halilunaziru73-creator>

## CONFERENCES, SEMINARS AND WORKSHOPS

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- Data Management Workshop: From Phenotyping Experiment to FAIR Data, Norway (2026)
- Digital Agroecology Cluster Conference (EU-funded), Brussels (2025)
- 6th African Regional Conference on Irrigation and Drainage, ARCID (2025)
- Vibrational Spectroscopy & Chemometrics in Agro-Food Systems (EU-funded) (2025)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, University of Florida (2025)
- IEEE Intl. Conf. on Smart Agriculture Technologies (AgriTech), Valencia, Spain (2024)
- NAERLS Technical Seminar Series: AI in the Food Value Chain (2023)

## CERTIFICATIONS AND DOCUMENTED RECOGNITIONS

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### Honorary Awards

- Award of Merit, Senator (SRC/SUG), Samaru College of Agriculture SUG (2017/2018)
- Certificate of Merit, Best 200 Level Student, Dept. of Agric. & Bio-Resources Engineering, ABU (2021)
- Best Overall Graduating Student, National Diploma, Samaru College of Agriculture (2018/2019)

### Training

- Certificate of Continuing Professional Development, ICCP, Envirofly Consulting UK (2024)
- Tony Elumelu Foundation Certificate, Business Management Training (2020)
- Genomics and Bioinformatics Bootcamp and short courses (4 certificates), Genomac Institute Inc., USA (2024–2025)
- SIWES Industrial Attachment Attestations, NAERLS and IAR, ABU (2017–2023)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, ISFOOD/UPNA seminar (2025)

### Recognition

- ASABE Member Certificate, Member No. 1061249 (2022)

- NIAE Student Member Certificate, Membership No. S2644 (2024)
- Conference Paper Acceptance, 6th ARCID (2025)
- IJRSI Publication Certificates (3), Vol. 13 Issue 7 (2026)
- SMEDAN Certificate of Registration, N.halilus Biogas Plant & Sells (2020)
- Attestation to Proficiency in English Language, ABU (2024)

#### **West African Examinations Council**

- West African Senior School Certificate (WAEC), Knowledge Is Power Secondary School, Zaria, 9 subjects including A1 in Mathematics and Geography (Jun 2015)

## **GALLERY AND PORTFOLIO LINKS**

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Field and campus photographs: [View gallery](#) | Full interactive portfolio: [halilunaziru73-creator.github.io](https://halilunaziru73-creator.github.io) | GitHub: [github.com/halilunaziru73-creator](https://github.com/halilunaziru73-creator)

## **REFERENCES**

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**Prof. Javier Casalí Sarasibar**, Public University of Navarre, Spain · [jcs@unavarra.es](mailto:jcs@unavarra.es)

**Prof. Konstantinos Soulis**, Agricultural University of Athens, Greece · [soco@aua.gr](mailto:soco@aua.gr)

**Dr. Ibrahim Babangida Dalha**, Ahmadu Bello University, Nigeria · [ibdalha@abu.edu.ng](mailto:ibdalha@abu.edu.ng)